Dynamic Pricing Recommendations

Dynamic Pricing Strategy — Recommendations & Implementation

The presentation focuses on increasing revenue per class (through optimized pricing. It draws directly from the EDA (demand patterns, high-value classes), the dynamic pricing (elasticity, forecasts, revenue projections), and the dataset (historical trends).

Executive Summary

- Based on the analysis of **3,271 historical classes**, demand modeling, and a price-elasticity-based dynamic pricing algorithm:
- **Key Conclusions**
- **Demand is INELASTIC (Elasticity = -0.2218)**
  → Prices can be increased significantly with minimal drop in bookings
- **High-demand classes (80%+ occupancy)** can absorb **20–30% price increases**
- **Morning (6–8 AM) & Evening (7 PM)** slots have consistently high occupancy
- **Weekend (Fri–Sun)** classes can support additional **10–20% pricing uplift**
- Dynamic pricing simulation shows **+20% to +35% total revenue uplift**

Understanding the Data

Dataset Highlights:
•3,271 rows, 13 features
•No missing values
•Mean occupancy ≈ **57.2%**, strong baseline demand

Categorical insights:
•Activity sites distributed across 5 key centers
•Demand highest Tue/Wed/Thu; lowest Sat/Sun (activity volume)
•But **utilization highest on Sunday**

Demand Behavior

⭐ **Day of Week Utilization**

- **Sunday highest** occupancy
- Monday & Thursday lowest
- Mid-week stable

⭐ **Hour-of-Day Demand**

- **Peak demand:**
    - **6 AM**
    - **8 AM**
    - **7 PM**
- **Lowest demand:** 2 PM–4 PM

⭐ **Seasonal Trend**

- Weeks 13→26 show **declining occupancy** → opportunity for seasonal dynamic pricing

High-Demand Classes

Classes repeatedly appearing with highest utilization:

- Aquafit (multiple sessions) — strongest performer
- Body Combat
- Pilates
- Zumba
- Step sessions

📌 **These are prime candidates for aggressive pricing increases.**

Dynamic Pricing Model Summary

Model uses:

- Forecasted occupancy
- Time-of-day demand patterns
- Day-of-week patterns
- Price elasticity integration
- Revenue simulation

Pricing multipliers:

- High demand (>80%) → +25%
- Peak hours (6–8 AM, 7 PM) → +5% to +15%
- Weekends → +10%

Price Elasticity Insight

From your dynamic pricing model:

- Elasticity = -0.2218
- **Meaning:**
- A 10% price increase → only 2.2% drop in demand
- A 20% increase → ≈4.4% drop
- A 30% increase → ≈6.7% drop
- 📌 **Interpretation:**
- Demand is strongly INELASTIC → customers accept price hikes.
  Therefore…
- ➞ **Revenue increases MORE from price hikes than it decreases from booking loss**.

Example Pricing Outcome

Example= 20-20-20 2.45pm-3.45pm (Activity)
Current price - ₹499

10% Price Increase:

- New Price: ₹549
- Demand Drop: 2.2%
- New Bookings: 11.7
- Revenue Increase: +7.6%

20% Price Increase:

- New Price: ₹599
- Demand Drop: 4.4%
- New Bookings: 11.5
- Revenue Increase: +14.7%

30% Price Increase:

- New Price: ₹649
- Demand Drop: 6.7%
- New Bookings: 11.2
- Revenue Increase: +21.3%

RECOMMENDATIONS:

1. INCREASE prices for popular classes (80%+ occupancy) by 20-30%
2. INCREASE prices during peak hours (6-9 PM) by 15-25%
3. INCREASE weekend prices by 10-20%
4. Monitor actual bookings to refine elasticity estimate

Strategy Comparison

Strategy	Effect	Revenue Impact	Total Revenue	Revenue Change
Current Fixed Price	Baseline	0%	₹1,694,258	0%
Dynamic Pricing (No Elasticity)	Raises prices only	positive	₹2,100,819	+24%
Wrong Elasticity (−1.2)	Too aggressive	Negative	₹1,420,444	-16.2%
Dynamic Pricing (Correct Elasticity)	Optimal	Highest Revenue Gain	₹1,977,201	+16.7%

Best performing strategy:
Dynamic Pricing with accurate elasticity → Highest revenue lift

Note- 'Dynamic Pricing (No Elasticity)' is MATHEMATICALLY IMPOSSIBLE It assumes demand doesn't change when prices change, which never happens in reality. With actual elasticity of -0.2218, use the 'Correct Elasticity' strategy.

Recommended Pricing Actions

1. **High-Occupancy Classes (80%+)**
   - Increase price by 20–30%
   - minimal drop in demand (<7%)
2. **Peak Hour Pricing**
   - Morning (6–8 AM): +5–10%
   - Evening (6–9 PM): +15–20%
3. **Weekend Pricing (Fri–Sun)**
   - Add 10–20% markup
   - Weekend demand is emotionally driven → less sensitive
4. **Low-Demand Slots (2–4 PM)**
   - Apply 10–15% discounts(Happy hours offer)
   - Boost occupancy, widen funnel
5. **Seasonal Adjustments**
   - Occupancy decline in weeks 13–26
   - Offer targeted promotions or small price reductions

Implementation Roadmap

Weeks 1–2

✓ Test 15–20% increase for top 10% classes
✓ Start with Aquafit, Body Combat, Pilates

Weeks 3–4

✓ Measure booking impact
✓ Re-calibrate elasticity estimate

Weeks 5–6

✓ Roll out time-based pricing (+15% evenings)

Weeks 7–8

✓ Implement weekend pricing (+10–15%)

Ongoing

✓ Weekly demand forecasting
✓ Monthly elasticity refresh
✓ Continuous optimization

EXPECTED RESULTS:

Month 1-2: +5-10% revenue increase

Month 3-4: +15-25% revenue increase

Month 5-6: +25-35% revenue increase (with optimization)